

Two tensor stages with compatible controlled corners

A conditional multiplication saving $15536/10^9 > 2^{-16}$

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With substantial OpenAI Codex assistance

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Abstract

We combine Paureel’s two-stage partial-swap topology, including its paid rank-one copy correction, with icekylinx’s retained-total producers and controlled inner basis. At factor dimensions (55, 53), one rational basis supports both auxiliary small corners and all local projector blocks. The resulting bit saving is $15537/10^9$. Keeping the PR21 complex circuit and PR23 semantic/bulk transfer gives the conditional final saving $\kappa = 15536/10^9 > 2^{-16}$. The construction uses actual regenerated finite producers, exact rational containment checks and an exact modular nonvanishing witness for the common-basis argument. It retains the predecessors’ analytic and fixed-tape hypotheses; it is not a formal verification or a practical multiplication implementation.

1 Dependencies and scope

The retained computational model is a fixed finite-alphabet multitape Turing machine. Every finite graph, rational basis, eligible odd prime, constant and tape count is independent of input length. Payload arithmetic in the new network is over $\mathbb{F}_2$; address geometry is rational, subsequently reduced at an eligible odd prime. These fields are distinct. The complex payload remains Gaussian dyadic.

The two-stage topology and its one-child correction are Aurel Prosz’s (Paureel) construction, pinned at `c82d09eb4781b68435e3fa3eb6beea72fa900fab`. Swapnil Jain’s `integer-mult-kappa` research led us to this topology and its macro batching. The ordinary-base-four producer and controlled inner basis are icekylinx PR24, pinned at `ed90fd940279c336ebc45968631bdebfb087b505`, building on PR18/10. The unchanged complex circuit is Zhihao Chen’s PR21; the semantic guard, product row stock and analytic composition are PR23, pinned at `661ebabc1076c9f525d7ce4967c876b203fa9827`. RaD/hipotures PR20 supplies the credited semantic, routing, phase-cell inverse and bulk ingredients, with the provenance and CC0 notices already preserved there.

The present contribution is the compatible unequal two-factor composition, its physical accounting, a common-basis nonvanishing certificate, the two regenerated finite graphs and the new assembled witness. Earlier credits, including eumemic’s source frames, Douglas Colkitt’s framework, OpenAI’s upstream manuscript, and Harvey–van der Hoeven’s analytic background, remain in force. PR25’s A5 subblock and PR27’s endpoint composition are concurrent work; neither extra block nor their numerical witness is used. Earlier GPT-6 Astra assistance remains attributed as recorded, without inferring the present runtime model. Future work using this contribution should acknowledge Zhihao Chen and every other dependency used. This is a provenance request, not an additional license condition or a worldwide priority claim.

2 Finite scalar interfaces

For $h = 53, 55$, let $v_h = \binom{h}{3}$ and use the rational metric $I - J/9$. A triple indicator has norm 2, and indicators with intersection one are orthogonal. The PR24 producer retains three side outputs per target and all h point totals. For triples S, T , their combined scalar coefficient is

$$[|S \cap T| = 1] + |S \cap T| \pmod{2} = [S = T].$$

Thus the designated injection J , reversible producer L and source copy V satisfy $JLV = I$. The chronological word $L, J, L^{-1}, V, L, J, L^{-1}, V$ adds the input to the target and restores every auxiliary, regardless of its initial contents. Both orientations use this same identity and reversed complementary frames.

The cancellation-free paired exclusion producer, equal-support sharing, positive labels and oriented carrier matching are unchanged algorithms from PR24/18, now run at the selected two dimensions. Their outputs are

h	v_h	c_h	q_h	$ \mathcal{M}_h $	R_h	ℓ_h
53	23426	518699	70331	8243	580787	2756
55	26235	582658	78760	8829	652589	2970

Here $R_h = c_h + q_h - |\mathcal{M}_h|$ already includes the retained centers; no additional h center roles are added. Their loss is $\ell_h = h(h - 1)$, not h^2 . The internal rank histograms satisfy

$$\sum_r r H_{h,r} = h R_h + 2 \ell_h.$$

They are supplied entry by entry in the producer certificates.

Every actual support and output was rebuilt. In addition to the inherited matching checks, a separate exact integer audit checks that each original support envelope lies in its enlarged positive label and that every dependency is a rational subspace inclusion. It checks 1,101,357 envelopes, 2,202,714 directed dependencies and 149,091 output frames. A positive label lies in $U_i = \{x : \sum_j x_j = 3x_i\}$, where $x^t(I - J/9)x = \sum_{j \neq i} x_j^2$; hence it is nondegenerate. Inclusion duality gives the reverse complementary orientation exactly. Carrier continuations are separately required to be increasing in event order and compatible in the matching algorithm. The retained proof of that compiler therefore applies to every generated gate and both invocation orientations.

3 Two stages, endpoints and the copy correction

Put $a = 55$, $b = 53$, $m = ab = 2915$, $N = v_a v_b$ and $\mathcal{F} = F_a \otimes F_b$. Write P, Q for the two triple-line projectors and $U = \text{im}(P \otimes Q)$. For any rational projector A , the partial-swap address frame is

$$D_A = \begin{pmatrix} I - A & A \\ A & I - A \end{pmatrix}.$$

Nested nondegenerate labels give $D_A D_B = D_{B-A}$ when $A \leq B$; the same positive residual executes either orientation.

Use the first forward scalar shear, followed by the second inverse shear. Their binary scalar map is $(x, y) \mapsto (y, x + y)$, with distinct restored auxiliary banks. The data source frames are (D_U, I) and terminal frames $(D_{\mathcal{F}}, D_{U^\perp})$. The tensor-stage boundary labels are the first two stages of the

retained construction: stage one grows $X : U \rightarrow F_a \otimes \text{im } Q$ and $Y : 0 \rightarrow \text{im}(I - P) \otimes \text{im } Q$; stage two ends at $(\mathcal{F}, U^\perp)$.

Stage-one auxiliary source and sink are $0, \mathcal{F}$. Its final outer residual is $I_a \otimes (I_b - Q)$, rank $m - a$. In stage two use auxiliary source $D_0 = (I_a - P) \otimes I_b$ and complementary sink $I - D_0$. Raise the physical data source to D_0 before its initial inverse copy, as in PR21/24. Every subsequent local label contains D_0 , so the auxiliary entrance vanishes. Cleanup ends at I , giving exit D_{D_0} , rank $m - b$. Arbitrary-scratch restoration makes the total auxiliary endpoint $D_{I-D_0} D_{D_0} = D_I$; no physical preparation or free reordering is assumed. The two stage banks are not shared.

Let $F = D_I$, $E = D_{I-P \otimes Q}$ and $T = D_{P \otimes Q}$. The framed data outputs are

$$A = Fy, \quad B = Fx + Ey.$$

Retain A , copy it to a temporary stream, apply T to that copy, and XOR into B . Since $TF = E$, the result is (Fy, Fx) . This is exactly Paureel's correction, charged as one rank-one child per pair. Its volume is the existing role volume V/W . Copy, XOR and erasure cost $O(V/W)$; the original is parked without recursively restoring it. Processing these copies sequentially adds a fixed stream to the retained parking stack, not a new row-index family. After undoing the data row-bank exchange, every role, including arbitrary scratch, receives full interchange.

4 One common basis for both axes

Use the PR24 inner family with physical index $i = b\alpha + \beta$,

$$K = T_2(I_a \otimes L_b), \quad T_2(x \otimes e_\beta) = M_\beta L_a x \otimes e_\beta.$$

Prescribe first- $2a$ row coefficient labels $R = [0, \dots, a-1, 1, \dots, a-1, 0]$ and last- $2a$ inverse-column labels $C = [a-1, 0, \dots, a-2, 0, \dots, a-1]$. Their effective residues are $\{0, 1, 3\}$ and $\{b-5, b-3, b-2\}$ modulo b . They are disjoint at $b = 53$; the finite certificate constructs all 53 permutation completions. Fix these M_β , leaving L_a, L_b free.

Absorb L_a, L_b into local matrices and denote the transformed line projectors by $p\xi, v\nu$, with $\xi p = \nu v = 1$. If boundary row i and column j have labels r_i, c_j and second-factor indices β_i, δ_j , the two local corner formulas are

$$(A \otimes v\nu)_{ij} = v_{\beta_i} \nu_{\delta_j} A_{r_i, c_j},$$

$$(p\xi \otimes B)_{ij} = p_{r_i} \xi_{c_j} B_{\beta_i, \delta_j}.$$

These are direct contractions of the prescribed rows and inverse columns. On first/last a coordinates, $r_i = c_i = i$. Thus the corner of $I_a \otimes v\nu$ is diagonal and invertible, and any local A is represented up to nonzero diagonal row and column scalings. On first/last b coordinates, $\beta_i = \delta_i = i$, $r_i = i$, $c_i = i + 2$. Thus $p\xi \otimes I_b$ also has an invertible diagonal corner, and every local B is represented up to such scalings. Consequently the two auxiliary profiles are $(a, m - 2a)$ and $(b, m - 2b)$ simultaneously.

For a local rank- r projector in dimension h , the retained generic projector identity gives $h - r$ singleton pivots and a block $2r - h$ if $2r > h$; otherwise retain r singletons. At $r = h$ use a whole h -block. The first/last local corner formulas just proved transfer all these profiles to the ambient physical order. Each needed ordinary minor is nonzero as a rational function of the corresponding free L_h by similarity of rank- r projectors and the usual rank-one pairing witness.

The remaining data macro is $(I - P) \otimes (I - Q)$, with nullity $d = a + b - 1 = 107$. Its null-corner has entries

$$Z_{ij} = [\beta_i = \delta_j] p_{r_i} \xi_{c_j} + [r_i = c_j] v_{\beta_i} \nu_{\delta_j} - p_{r_i} \xi_{c_j} v_{\beta_i} \nu_{\delta_j}.$$

The exact certificate supplies normalized line coordinates modulo the prime 1,000,003 and proves this 107-square matrix has full rank. This is a nonvanishing witness, not a numerical inference: lift the coordinates to integers and normalize the two dual vectors over $\mathbb{Q}$. The normalization denominators are 1 modulo this prime, so the lifted rational determinant is nonzero. Any fixed pair of rank-one projectors can be conjugated over $\mathbb{Q}$ to these lifted projectors. Hence this determinant is a nonzero rational function on the free $\mathrm{GL}_a \times \mathrm{GL}_b$ family for each physical data pair. The large-projector identity then gives 107 singleton pivots and a block $m - 214 = 2701$.

There are finitely many physical local projectors and data pairs. The ordinary minors, line coordinates and displayed data determinants are individually nonzero rational functions on this same irreducible parameter space. Clearing denominators, their finite product is nonzero. Rational enumeration therefore constructs one common basis satisfying all requirements. Choose the address prime after this choice, avoiding its finitely many bad denominators and minors. The witness prime above is only a certificate device and need not be this address prime.

The retained lower-triangular conjugation for D_A turns every contiguous identity-pivot block into one child interchange. Nonzero diagonal scales are absorbed in the triangular factors. Thus there is no uncharged gather, reversal or independent per-edge choice of basis.

5 Complete physical list and exact moment

Set $B_a = v_b R_a$, $B_b = v_a R_b$,

$$W = 2N + B_a + B_b = 31753659079, \quad L = v_b \ell_a + v_a \ell_b = 141878880.$$

The following list includes all physical edges and the correction:

class	multiplicity	profile
first auxiliary exterior	B_a	55, 2805
second auxiliary exterior	B_b	53, 2809
interstage data macro	$2N$	107 singletons; 2701
axis h internal rank r	$v_{\text{other}} H_{h,r}$	ordinary (h, r)
axis 55 physical growth	$2N$	1, 53
axis 53 physical growth	$2N$	1, 51
endpoint copy correction	N	1

The local histograms exclude auxiliary exterior and physical data edges; their rank identity therefore checks this enumeration without overlap. Both orientations have the same positive residual profiles. Its rank is

$$s = Wm - N + 2L = 92561585391935, \quad Wm - s = 330823350.$$

The extra N correction calls are already included. Each child has the same role-volume fraction $1/W$, including the copied stream.

Let n_t be this exact list, $a_b = 15537/10^9$ and $\tau = 1 - a_b$. The retained rational logarithm enclosure and $e^u \leq 1 + u + u^2/[2(1 - u/3)]$ give

$$\frac{\sum_t n_t t^\tau}{W m^\tau} < 1 - 2.7 \times 10^{-10}.$$

All widths satisfy $0 < t < m$; the largest is 2809. The complete-row mixed-width proof from PR10/18 consequently gives the arbitrary-width bit interchange interface, including remainders, compact spectators and the depth-first fixed-tape schedule. The correction acts on a complete copied role stream with exactly these same compact/spectator ranges, so its cost and divisibility obey that proof as well.

6 Semantic precision, row stock and final assembly

The PR21 complex graph and all its physical phase operators are unchanged: $a_c = 18/10^6$, $m_c = 21952$, $W_c = 2085111546336$ and maximum child 21924. PR23's literal complex scalar charge, fixed fine-grid format, completed-child operator bound, and resulting $C_1 = 1$ therefore remain valid. The new bit primitive supplies the identical arbitrary-width contract to its exact address/record adapters; it changes their time exponent, not Gaussian arithmetic or any child operator. No complex path guard is transferred from a different graph.

The bit halving degree is 19 and its wire exponent is 35; the complex values remain 544 and 41. The product row-stock coefficient is $19 \cdot 35 + 544 \cdot 41 = 22969$. Since $47000 > (51/25)22969$, a p^{47000} stock suffices by the same PR23 product argument. Use one leading prefix, one padding, and restore the bit factor within every complex descendant. A parked correction copy introduces no new independent row index or internal padding.

Choose $a = a_b$, $b = a_c$, $h = 10^{-8}$ and $\beta = 1/10$, and set

$$\begin{aligned} q &= a(1 - 2h), \quad \lambda' = 1 - q, \quad \lambda = (1 - a + \lambda')/2, \quad c = q(1 + h), \\ \epsilon &= \frac{1 - h}{1 + c + q}, \quad G = \epsilon q, \quad r = (G + 1 - \epsilon)/2, \quad \delta = h/8, \quad C_1 = 1. \end{aligned}$$

The complex leaf saving is $(1 - \beta)b = 16.2 \times 10^{-6} > a$. PR23's exact seven margins are

$$1 - \epsilon(1 + c), \quad a, \quad G, \quad a, \quad \min(1 - \epsilon - \delta, r - \delta), \quad 1 - \epsilon - \delta, \quad \epsilon.$$

The certificate recomputes all 47 strict conditions, including the actual scalar charge and new product-stock gap. Their minimum margin is G , and

$$G - \frac{15536}{10^9} > 5 \times 10^{-10}.$$

The strict slack absorbs logarithmic factors. The unchanged arbitrary-source router, phase-cell inverse, microbox halo and fixed-tape bulk interfaces, prime-packing argument, and exact-recovery conditions of PR23 apply at these parameters. Their fixed constants and setup, logarithmic absorption, prime thresholds and table domination remain eventual conditions; no practical input-size cutoff is asserted here.

Theorem 1. *Under the retained analytic and fixed-tape interfaces, the construction gives*

$$T(n) = O(n(\log n)^{1-\kappa}), \quad \kappa = \frac{15536}{10^9} = 1.5536 \times 10^{-5} > 2^{-16}.$$

Proof. The finite scalar identities, label inclusions, physical list, common basis and charged correction establish the bit interface. The complex interface and semantic bound are retained by source hash. The product stock supplies both recursions, and the exact margins above instantiate the PR23 assembly. $\square$

7 Reproduction and verification boundary

`python3 research/two-stage/producer.py` regenerates both new graphs, all scalar supports, labels, matchings and histograms, and the separate integer containment audit. `python3 research/two-stage/verify.py` checks the exact moment, modular common-basis witness, retained source hashes, row stock and final assembly. Four focused tests include the complete small

data/dirty-auxiliary endpoint basis, missing-correction and wrong-order negatives, and rejection of the old guard/exposure formulas and 2^{-15} . The unchanged PR23 producer and analytic controls are preserved by source hash rather than unnecessarily rerun. Finite evidence and written proofs have distinct scopes; this is neither full formal verification nor completed external expert review. The canonical citations and source licenses are listed in `research/two-stage/SOURCE.json` and NOTICE.